

OptiCrop: Smart Agricultural Production Optimization Engine

OptiCrop is a machine learning-based crop recommendation system. Given soil and climate readings (nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall), it predicts the most suitable crop to grow, and includes exploratory analysis to group crops by season and by soil/climate profile.

Problem Statement

Recommend the best crop for a given set of soil and environmental conditions, to support data-driven farming decisions and improve agricultural productivity.

Approach

- A Logistic Regression classifier trained on standardized soil/climate features, evaluated against 22 crop classes.
- Rule-based seasonal grouping (Kharif / Rabi / Summer) for interpretable, agronomically meaningful crop categorization.
- K-Means clustering to uncover natural soil/climate groupings independent of crop labels.
- A Flask web application serving the trained model for real-time predictions via a form-based UI.

Model Results

Metric	Score
Accuracy	97.27%
Precision	97.42%
Recall	97.27%
F1-Score	97.25%
ROC-AUC	0.9998

K-Means Clustering Findings

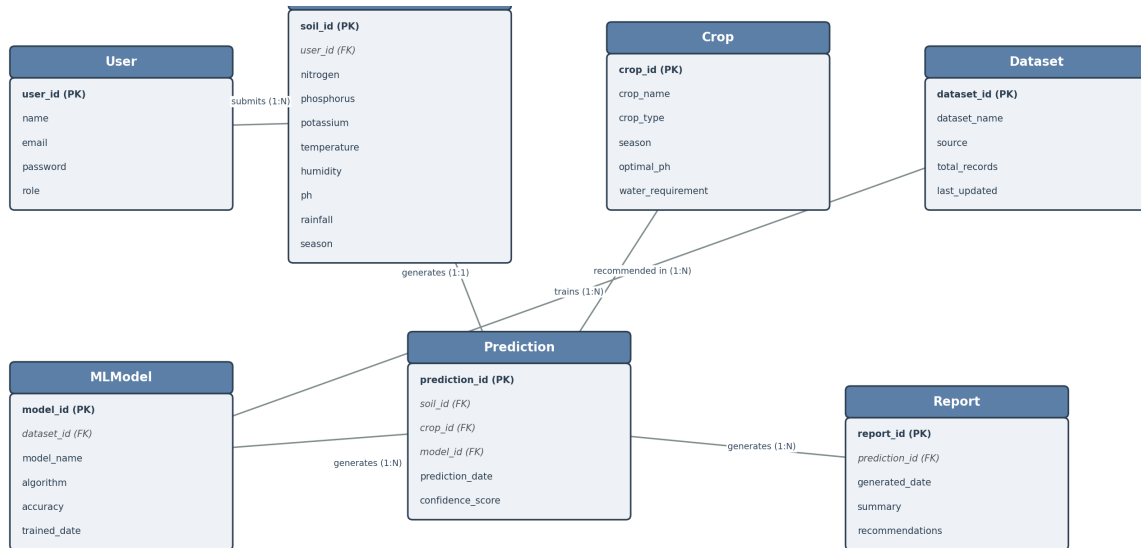
K-Means ($k=4$, chosen via the elbow method) revealed four natural soil/climate clusters independent of crop label: a moderate-nitrogen/high-phosphorus/low-rainfall cluster, a moderate-nutrient/high-humidity/high-rainfall cluster, a high-nitrogen/moderate-phosphorus cluster, and a low-nitrogen/very-high-phosphorus/high-rainfall cluster with lower pH. These groupings suggest the dataset naturally organizes around distinct soil-nutrient and moisture profiles.

Repository Structure

app.py (Flask app) · Crop_recommendation.csv (dataset) · model/ (trained artifacts) · notebook/ (full analysis pipeline) · templates/ & static/ (frontend) · requirements.txt

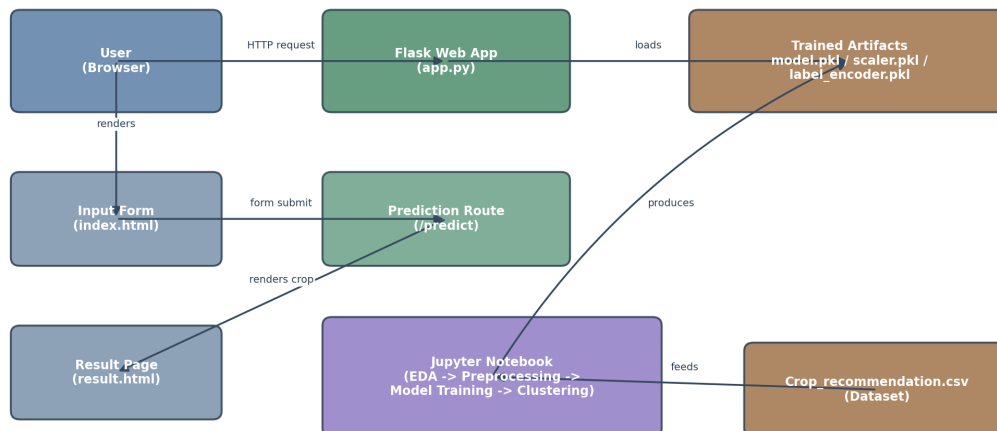
Entity Relationship Diagram

OptiCrop — Entity Relationship Diagram



System Architecture

OptiCrop — System Architecture



Future Improvements

- Compare Logistic Regression against ensemble models (Random Forest, XGBoost).
- Incorporate soil type or geographic region as additional features.
- Deploy the model behind a REST API for integration with other tools.